

Extending Theorem 2 of *Robust Trust* to infinite M and Θ : An exposition of what we proved, what is tight, and what remains open

Pedro Aldighieri

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Abstract

This note reports on a proof, drafted over the past two days, of a two-tier conditional infinite-dimensional extension of Theorem 2 (“Robustly Rationalizable Solution”) from Dworczak & Smolin’s *Robust Trust* (2026). Existence of an optimal value-securing agent strategy σ^* is established under the paper’s standing hypotheses plus one new endogenous regularity condition (**A5-thick**). Adversary attainment $U(\beta^*, \sigma^*) = U^*$ requires one more structural condition (**A8c-attain**); full Definition 2 robust rationalizability requires a third, posterior-calibrated transport condition (**TRE-gen-Hall**). All three added hypotheses are *tight*: explicit obstruction witnesses rule out their removal in general. The route follows Phil Reny’s restricted-game-plus-Lusin-lift suggestion, with a number of detours that the original sketch does not spell out. Below is the structure of the argument and the boundaries.

1 The setting and the question

We use the notation of *Robust Trust*. Thus Ω is finite with full-support prior μ_0 ; $s \in \Delta(\Omega)$ has state-conditional law $\pi(\cdot \mid \omega)$ and unconditional law $\tau = \sum_{\omega} \mu_0(\omega) \pi(\cdot \mid \omega)$; $M = \text{supp}(\tau)$; $\theta \in \Theta$ (compact metric) is drawn from $f(\cdot \mid \omega)$, independently of s conditional on ω ; A is compact metric; and $u(a, \omega, \theta)$ is bounded and continuous in a . Let Σ denote the agent’s measurable strategies $\sigma : M \times \Theta \rightarrow \Delta(A)$ and B the misaligned adviser’s measurable kernels $\beta : M \rightarrow \Delta(M)$. With probability α the adviser is aligned and reports s truthfully; with probability $1 - \alpha$ the adviser is misaligned.

The agent’s robust objective is

$$U(\sigma) = \alpha \mathbb{E}_{\text{id}, \sigma}[u] + (1 - \alpha) \inf_{\beta \in B} \mathbb{E}_{\beta, \sigma}[u], \quad U^* = \sup_{\sigma \in \Sigma} U(\sigma), \quad (1)$$

and Definition 2 (*robust rationalizability*) asks for an agent strategy σ^* together with an adversarial β^* such that, at every on-path message m , the agent’s prescribed continuation is Bayes-optimal under the posterior $P_{\beta^*}(\cdot \mid m)$.

The paper’s Theorem 2 establishes existence of such a (σ^*, β^*) under the additional assumption that M and Θ are *finite*, via a finite Sion saddle. The paper itself flags the difficulty of extending the saddle argument to infinite cheap-talk-like spaces. The question addressed here is whether the existence direction of Theorem 2 can be extended when M and Θ are infinite.

2 Main theorem (two tiers, with three added hypotheses)

The standing hypotheses of *Robust Trust* are kept verbatim (Ω finite, μ_0 full support, A, Θ compact metric, u bounded continuous in a , conditional independence of s, θ given ω , all measurability

Borel). Three additional hypotheses enter the extension. Section 6 explains the obstruction attached to each.

Assumption A5-thick (endogenous Lusin-thickness). There exist compact $K_1 \subseteq K_2 \subseteq \dots \subseteq M$ with $K^* := \bigcup_n K_n$ such that $\pi(K^* \mid \omega) = 1$ for every ω ; the Branch-A maximizer's representative $\hat{\sigma}^*$ is continuous on each K_n in the Balder stable private-strategy topology; and every relative open in K_n has positive $\pi(\cdot \mid \omega)$ -mass for every ω .

Assumption A8c-attain (rowwise argmin attainment). For τ -a.e. s , the rowwise argmin $D(s) := \arg \min_{m \in M} \ell_{\sigma^*}(m, s)$ is nonempty, where $\ell_{\sigma^*}(m, s) := \sum_{\omega} s(\omega) p_{\omega}(m)$ with $p_{\omega}(m) := \int_{\Theta \times A} u(a, \omega, \theta) \hat{\sigma}^*(m, \theta)(da) f(d\theta \mid \omega)$, and $s \mapsto D(s)$ admits a Borel measurable selector m^* .

Assumption TRE-gen-Hall (calibrated worst-message transport). The Branch-A maximizer σ^* has trust-region structure: closed convex $T \subseteq \Delta(\Omega)$, continuous Bregman projection P_T , $\hat{\sigma}^*(m)$ Bayes-optimal at $P_T(m)$, with m^* the selector from A8c-attain (now also interpreted as the single-valued monotone worst-message map). Moreover the joint coupling $\gamma_{\alpha} := \alpha(\text{id}, \text{id})_{\#} \tau + (1 - \alpha)(\text{id}, m^*)_{\#} \tau$ on $M \times M$, disintegrated along its second coordinate to give $\kappa(\cdot \mid m)$, satisfies the posterior-calibration condition $P_{\gamma_{\alpha}}(\cdot \mid m) \in C(m)$ for q -a.e. m , where $q := (\gamma_{\alpha})_2$ and $C(m) := \{\mu \in \Delta(\Omega) : \hat{\sigma}^*(m) \in \arg \max_{\hat{\sigma}'} U(\hat{\sigma}', \mu)\}$ (the set of beliefs at which $\hat{\sigma}^*(m)$ is Bayes-optimal).

Equivalently (support-function form): for every measurable $E \subseteq M$ and every continuous affine $\phi : \Delta(\Omega) \rightarrow \mathbb{R}$,

$$\alpha \int_E \phi(m) \tau(dm) + (1 - \alpha) \int_{(m^*)^{-1}(E)} \phi(s) \tau(ds) \leq \int_E h_{C(m)}(\phi) q(dm),$$

where $h_{C(m)}(\phi) := \sup_{\mu \in C(m)} \phi(\mu)$.

Theorem (Tier 1: optimal σ^* and adversarial β^*). Under the standing hypotheses and Assumptions A5-thick and A8c-attain, there exist $\sigma^* \in \Sigma$ and $\beta^* \in B$ such that

$$U(\sigma^*) = U^* = \sup_{\sigma \in \Sigma} U(\sigma), \quad U(\beta^*, \sigma^*) = \inf_{\beta \in B} U(\beta, \sigma^*) = U^*.$$

Theorem (Tier 2: robust rationalizability). Under the Tier 1 hypotheses plus TRE-gen-Hall, the pair (σ^*, β^*) can be chosen so that, additionally, when $\alpha > 0$,

$$\hat{\sigma}^*(m) \in \arg \max_{\hat{\sigma}'} U(\hat{\sigma}', P_{\beta^*}(\cdot \mid m)) \quad \text{for } \tau\text{-a.e. } m \in M.$$

Hence σ^* is robustly rationalizable in the paper's a.e./on-path sense.

3 Proof strategy in one paragraph

The proof follows Phil Reny's two-stage route. First restrict the misaligned adviser to τ -dominated kernels $\beta_{\varphi}(dm \mid s) = \varphi(m \mid s) \tau(dm)$, with φ in the convex set F of jointly measurable normalized densities. On this restricted game, Balder's constant-marginal continuity [2] makes $U_F(\sigma, \varphi) := U(\beta_{\varphi}, \sigma)$ continuous in σ under the Balder weak topology, and Σ is compact after the

usual Balder quotient. Mertens's asymmetric minmax theorem [3] then gives a restricted-game maximizer σ^* . Second, A5-thick lets us use Lusin regularization to smooth any unrestricted $\beta \in B$ into a τ -dominated $\varphi_\varepsilon \in F$ with only ε -loss at σ^* . Hence the restricted maximizer also secures the unrestricted value.

Branch A gives value security, not a full saddle. Adversary attainment needs A8c-attain. Per-message Bayes-optimality needs TRE-gen-Hall, because Branch A plus attainment does not imply the upper saddle inequality that finite Sion gives in the paper.

4 Branch A: existence of the optimal σ^* (under A5-thick)

4.1 The stage-1 topology

Set $X := M \times \Theta$, $\bar{f} := \sum_\omega \mu_0(\omega) f(\cdot | \omega)$, $\lambda := \tau \otimes \bar{f}$. Endow Σ with the Balder weak topology T_λ : $\sigma_n \rightarrow \sigma$ iff $\int g d(\sigma_n \lambda) \rightarrow \int g d(\sigma \lambda)$ for every Carathéodory test $g(x, a)$ measurable in x , continuous in a , $L^1(\lambda)$ -dominated. Working modulo λ -a.e. equality (the Balder quotient) makes Σ a compact convex Hausdorff locally convex space [2, §2 Thm. 2.3(a)].

Lemma 1 (L1 — constant-marginal continuity). *For every $\varphi \in F$, the map $\sigma \mapsto U_F(\sigma, \varphi)$ is T_λ -continuous.*

Sketch. Decompose the payoff into the aligned and the misaligned terms; both integrate Carathéodory tests against *fixed* marginals ($\pi(\cdot | \omega) \otimes f(\cdot | \omega)$ and $\tau \otimes f(\cdot | \omega)$ respectively). Apply Balder [2, Thm. 2.2, p. 268]. The misaligned term's L^1 -domination uses $r_\omega^\varphi(m) := \int_M \varphi(m | s) \pi(ds | \omega) \in L^1(\tau)$, which holds by $\pi(\cdot | \omega) \ll \tau$ (automatic from full-support μ_0) and Tonelli; no boundedness restriction on φ is needed. $\square$

Lemma 2 (L2 — compactness of Σ). *(Σ, T_λ) is compact convex; every simultaneous ω -marginal limit is induced by a single common kernel.*

Sketch. Compactness via Balder [2, §2 Thm. 2.3(a)], applied with finite base λ and compact metric target A . Common-kernel extraction: form the finite mixture $Q := \sum_\omega \mu_0(\omega) Q_\omega^\tau$ of any ω -indexed limits, disintegrate on the standard Borel space $X \times A$, and recover each ω -indexed marginal via Radon-Nikodym multiplications by the densities $d\lambda_\omega^\tau/d\lambda$ and $d\pi(\cdot | \omega)/d\tau$ (directions: small over large; both bounded by $\mu_0(\omega)^{-1}$). $\square$

4.2 Restricted-game maximin and Lusin lift

Lemma 3 (L3 + L4 — Mertens minmax + restricted maximizer). *With Σ compact Hausdorff in T_λ , F convex with no topology, and U_F continuous in σ (Lemma 1) and bounded by $\|u\|_\infty$, Mertens's Corollary B [3, §2, p. 238] gives $\max_\sigma \inf_F U_F = \inf_F \max_\sigma U_F = V^*$. Both mixed-strategy sides collapse to pure strategies by affineness: finite-support F -mixtures barycenter back into F , and agent-side Borel mixtures barycenter back into the compact convex Σ . Upper-semicontinuity of $V(\sigma) := \inf_\varphi U_F(\sigma, \varphi)$ plus compactness of Σ deliver a maximizer $\sigma^* \in \Sigma$.*

Lemma 4 (L5 — Lusin-thick compacts (A5-thick)). *Under A5-thick, the compact sequence $K_n \uparrow K^*$ exists with the three required properties: full $\pi(\cdot | \omega)$ -measure of K^* , Balder continuity of $\hat{\sigma}^*$ on each K_n , and support-thickness in K_n .*

Lemma 5 (L6 — Lusin lift). *For every $\beta \in B$ and every $\varepsilon > 0$, there exists $\varphi_\varepsilon \in F$ with $U(\beta, \sigma^*) \geq U_F(\sigma^*, \varphi_\varepsilon) - \varepsilon$.*

Sketch. Build a smoothing kernel $q_\varepsilon(z \mid y)$ shell-by-shell on $D_n := K_n \setminus K_{n-1}$, supported in $K_n \cap B(y, \rho_n)$ with positive τ -mass denominator from the support-thickness clause of A5-thick. Compose with β via Tonelli to obtain $\varphi_\varepsilon(z \mid s) := \int q_\varepsilon(z \mid y) \beta(dy \mid s)$. Pointwise uniform-continuity of p_ω on each K_n (from $\hat{\sigma}^*$ -continuity, which is one of the three A5-thick clauses) gives the bound $|U(\beta, \sigma^*) - U_F(\sigma^*, \varphi_\varepsilon)| \leq (1 - \alpha)\varepsilon$. $\square$

4.3 Branch A capstone

Theorem 6 (Branch A). *Under the standing hypotheses of Robust Trust and Assumption A5-thick, there exists $\sigma^* \in \Sigma$ with $U(\sigma^*) = U^* = V^*$ and $\inf_{\beta \in B} U(\beta, \sigma^*) \geq U^*$.*

Proof. $V^* \geq U^*$ since $F \hookrightarrow B$ via β_φ (the inf over a smaller set is larger pointwise in σ). Lemma 5 gives $\inf_B U(\cdot, \sigma^*) \geq V^*$, so $U^* \geq V^*$. Combining, $V^* = U^* = U(\sigma^*) = \inf_B U(\beta, \sigma^*)$. $\square$

This is genuinely the half of Theorem 2 that Phil Reny’s sketch was aiming at: *existence of an optimal value-securing agent strategy*. The remaining half — adversary attainment plus per-message Bayes-optimality — is taken up in Branch B.

5 Branch B: adversary attainment and Bayes-optimality

5.1 The dual value formula and rowwise contact

Lemma 7 (L8a — restricted dual value). $\inf_F U_F(\sigma^*, \varphi) = \text{const} + (1 - \alpha) \int_M e(s) \tau(ds)$ where $e(s) := \text{essinf}_m \ell_{\sigma^*}(m, s)$.

The lower bound is row-by-row pointwise; the upper bound is realized by $\varphi_n(m \mid s) := \mathbf{1}_{A_n(s)}(m) / \tau(A_n(s))$ for $A_n(s) := \{m : \ell(m, s) \leq e(s) + 1/n\}$.

Lemma 8 (L8c-Half-1 — pointwise inf = essential inf τ -a.e.). $\inf_m \ell(m, s) = \text{essinf}_m \ell(m, s)$ for τ -a.e. s .

Sketch. Let $a(s) := \inf_m \ell(m, s)$ and $e(s) := \text{essinf}_m \ell(m, s)$, so $a \leq e$ pointwise. Jankov–von Neumann selection (with the standard τ -a.e. Borelization) gives a measurable $d_n : M \rightarrow M$ with $\ell(d_n(s), s) \leq a(s) + 1/n$, hence $\inf_B U(\cdot, \sigma^*) - \text{const} = (1 - \alpha) \int a d\tau$. Lemma 7 plus Lemma 5 (bottom-density: $\inf_B = \inf_F$) give $\int a d\tau = \int e d\tau$. Combined with $a \leq e$: $a = e$ τ -a.e. $\square$

Lemma 9 (L8 — β^* rowwise adversarial under A8c-attain). *Under A8c-attain, the Borel selector m^* exists, and $\beta^*(dm \mid s) := \delta_{m^*(s)}(dm)$ satisfies $U(\beta^*, \sigma^*) = U^*$.*

This closes Tier 1. Note that the natural sufficient routes to A8c-attain are primitive: (P1) the agent’s Bayes-action correspondence is upper hemicontinuous with closed compact values, (P2) the trust-region projection is continuous and $\hat{\sigma}^*$ plays the Bayes-action at $P_T(m)$, or (P3) σ^* admits a closed-graph representative. Each of (P1)–(P3) implies A8c-attain via the standard measurable maximum theorem [4, Thm. 18.19].

5.2 The L9 saddle gap and the calibrated transport

Branch A plus Lemma 9 do not produce a saddle. What is delivered is the lower saddle inequality $U(\beta^*, \sigma^*) = \inf_\beta U(\beta, \sigma^*) = U^*$; what is *not* delivered is the upper saddle inequality $U(\beta^*, \sigma) \leq U(\beta^*, \sigma^*)$ for every σ . The standard route to per-message Bayes-optimality (the

third clause of Definition 2) goes through the upper saddle: if $\hat{\sigma}^*$ failed Bayes-optimality on a positive-measure set of messages, a Kuratowski–Ryll–Nardzewski selection of an improving private strategy would beat $\hat{\sigma}^*$ against β^* , contradicting the upper saddle. Without the upper saddle the contradiction never closes. Phil’s email itself acknowledges this: “*If correct, this establishes the existence of an optimal strategy for player 1, but not for player 2.*”

The paper’s finite proof gets the upper saddle for free, via Sion on finite-dimensional simplex products. We substitute it with TRE-gen-Hall.

Lemma 10 (L9b — calibrated transport gives Bayes-optimality). *Under the hypotheses of Theorem ?? and Assumption TRE-gen-Hall, the β^* from Lemma 9 satisfies $P_{\gamma_\alpha}(\cdot \mid m) \in C(m)$ for q -a.e. m , hence*

$$\hat{\sigma}^*(m) \in \arg \max_{\hat{\sigma}'} U(\hat{\sigma}', P_{\beta^*}(\cdot \mid m)) \quad \text{for } q\text{-a.e. } m, \text{ and (since } q \geq \alpha\tau \text{ when } \alpha > 0) \text{ for } \tau\text{-a.e. } m.$$

The proof is the substantive content of TRE-gen-Hall: posterior calibration on $C(m)$ is exactly the per-message Bayes-optimality condition, and the displayed γ_α is the explicit construction. For Ω binary, the posterior-calibration condition reduces to a single 1-dimensional mass-balance equation whose unique solution is the quantile transport in Appendix A.6 of *Robust Trust*; the analog generalizes to ternary radial cases via Appendix A.10 (per-direction quantile transports plus radial Bregman monotonicity).

6 Why each added hypothesis is needed (and tight)

We now spell out why none of the three added hypotheses can be dropped in the conclusions as stated.

6.1 A5-thick is tight: perfect revelation

If $\Omega = \{0, 1\}$ and the posteriors are perfectly revealing ($\pi(\cdot \mid \omega) = \delta_{\delta_\omega}$), then $M = \{\delta_0, \delta_1\}$ is two points and any compact K^* with full $\pi(\cdot \mid \omega)$ -measure for both ω must contain both points. The relative open $\{\delta_0\}$ inside $K^* = M$ has zero $\pi(\cdot \mid 1)$ -mass, so the support-thickness clause cannot hold on any full-measure compact core. Lemma 4 fails and the smoothing kernel of Lemma 5 cannot be built. *Some* endogenous thickness condition is necessary.

6.2 A8c-attain is tight: the upward-spike row obstruction (Q1)

Consider the row payoff $g(m) = m$ for $m \in (0, 1]$ and $g(0) = 1$. Then $\inf_m g(m) = 0 = \text{essinf}_m g(m)$ (with τ Lebesgue) but the argmin set $\{m : g(m) = 0\}$ is empty: a sequence $m_n \rightarrow 0$ achieves $g(m_n) \rightarrow 0$ but $g(0) = 1$. No probability measure on $[0, 1]$ attains the value 0: if the measure gives no mass to $\{0\}$, then it is supported on $(0, 1]$, where $g(m) = m > 0$, hence the integral is strictly positive; if it gives positive mass to $\{0\}$, the atom contributes $g(0) = 1$ to the integral.

This is the standard attainment obstruction for non-l.s.c. rowwise objectives: without some rowwise argmin condition, exact Borel-kernel attainment is not available. The Robust-Trust-compliant primitive realization where this row arises as ℓ_{σ^*} for the Branch-A maximizer is more delicate to certify (the strategy has to genuinely play badly at a single τ -null message), but no positive attainment theorem exists either.

The honest endpoint is: Tier 1 must keep A8c-attain (sufficient via any of the primitive routes (P1)–(P3) above), or the conclusion must be weakened to the unconditional ε -*approximate adversary* statement

$$\forall \varepsilon > 0, \exists \beta_\varepsilon \in B : U(\beta_\varepsilon, \sigma^*) \leq U^* + \varepsilon,$$

which holds free of A8c-attain by the Branch-A bottom-density alone.

6.3 TRE-gen-Hall is tight: a ternary non-radial separating witness (Q2)

For $\Omega = \{0, 1, 2\}$, $A = \{a_0, a_1, a_2\}$ discrete with payoffs $u(a_\omega, \omega) = 1$, $u(a_i, \omega) = -1$ for $i \neq \omega$, atomless full-support τ on $\Delta(\Omega)$, and a non-radial trust region $T = \{\mu : \mu(0) \leq 0.4\}$, the TR-strategy is well defined and the posterior-Bayes-optimality cones $C(m)$ are half-spaces in $\Delta(\Omega)$. Take $E = \{t_0\}$, a singleton boundary point of T , and the affine test $\phi(\mu) = \mu_1 - \mu_0$. Direct computation (see the Q2 prover log) gives Hall LHS = $(1 - \alpha)/9 > 0$ but RHS = 0.

The geometric obstruction is genuinely multi-dimensional. In binary state, each boundary message has only a single scalar direction to balance, and Appendix A.6 supplies the corresponding quantile transport. In the ternary case, the fiber collapsing into the boundary point t_0 has a 2-dimensional barycenter that points into the wrong plurality cone; there is no one-dimensional monotone transport that can absorb the vector imbalance.

The honest endpoint is: Tier 2 must keep TRE-gen-Hall, unless an additional structural condition is imposed — radial symmetry of T , zonotopal/orthant alignment, or a separability condition on the Bayes-optimal cones. The ternary radial case (Section 5.2 plus Appendix A.10 of *Robust Trust*) does verify TRE-gen-Hall via per-direction 1-dimensional transports.

7 Remaining directions

Three directions remain open that we did not close in either pipeline:

1. **A constructive Branch-A representative making A8c-attain automatic.** Even though A8c-attain cannot be removed in general, a more refined choice of Branch-A representative — one that builds in upper-hemicontinuity or closed-graph structure of the strategy correspondence, rather than choosing an arbitrary representative — would make A8c-attain automatic in the constructed σ^* . The L5 modification is already one move in this direction; a more systematic construction is open.
2. **Multi-dim Hall via additional structural conditions on T .** Identifying the largest class of geometric structures on the trust region for which TRE-gen-Hall holds — analogous to the binary Appendix A.6 case — is open. Candidates: separable trust regions, zonotopal T , trust regions invariant under a transitive group action.
3. **Formalization.** A faithful formalization would need to encode the constructive Balder/Mertens/Lusin route rather than the earlier abstract saddle-point stub.

Provenance

Proof logs (intermediate prover outputs, reviewer verdicts, counterexample constructions) are committed in the proof repository under `packets/` and `logs/`.

References

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